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Kim

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(54) **CONTROLLER AND MEMORY SYSTEM**

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See application file for complete search history.

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(57)

ABSTRACT

A controller includes an Error Correction Code (ECC) encoder adding a first parity to data to generate a data set, and encoding the data set to generate a first parity data set, a buffer temporarily storing the first parity data set, an ECC decoder decoding the first parity data set received from the buffer to generate a decoded data set, a first checker performing a Low Density Parity Check (LDPC) encoding on the decoded data set to generate an LDPC data set to which a second parity is added, and a second checker performing a syndrome check operation on the LDPC data set including the first and second parities.

17 Claims, 6 Drawing Sheets

Related U.S. Application Data

(60) Continuation of application No. 17/851,130, filed on Jun. 28, 2022, now Pat. No. 11,953,990, which is a continuation of application No. 17/307,129, filed on May 4, 2021, now Pat. No. 11,403,174, which is a division of application No. 16/555,264, filed on Aug. 29, 2019, now Pat. No. 11,099,932.

Foreign Application Priority Data

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(51) **Int. Cl.**

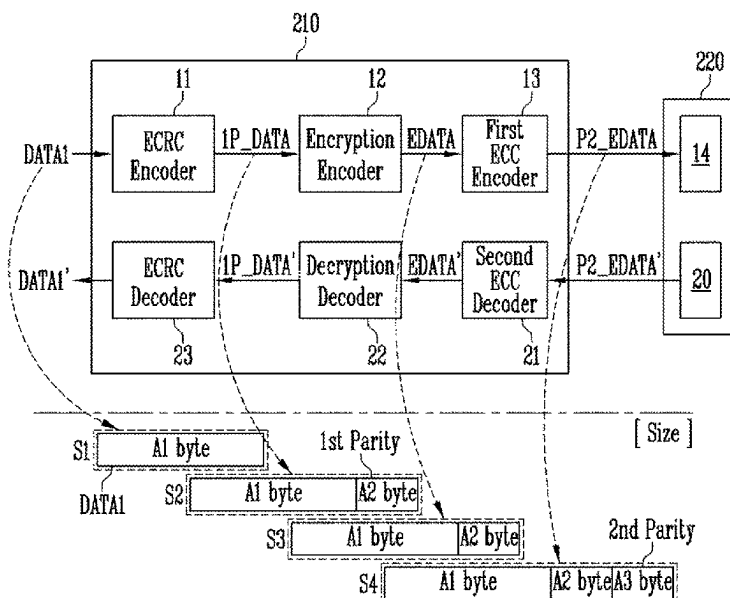
G06F 11/10 (2006.01)

G06F 21/60 (2013.01)

H03M 13/11 (2006.01)

(52) **U.S. Cl.**

CPC **G06F 11/1076** (2013.01); **G06F 21/602** (2013.01); **H03M 13/1102** (2013.01)



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FIG. 1

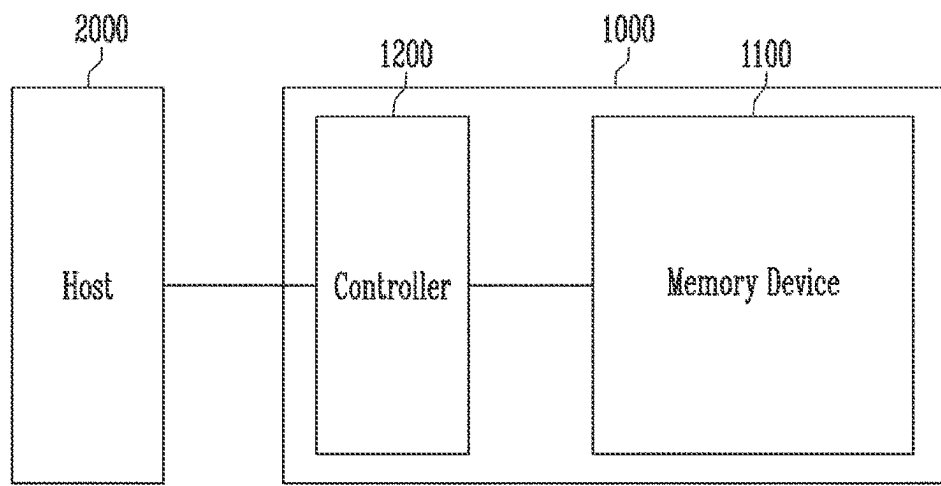


FIG. 2

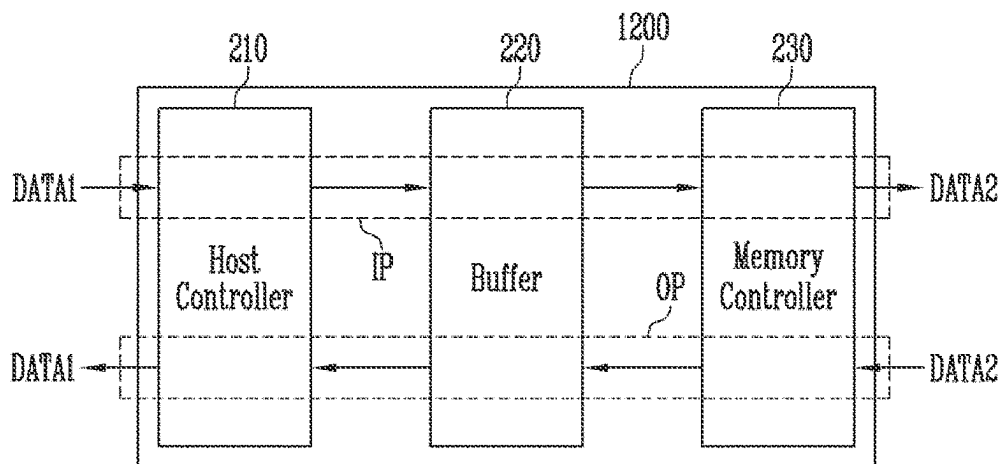


FIG. 3

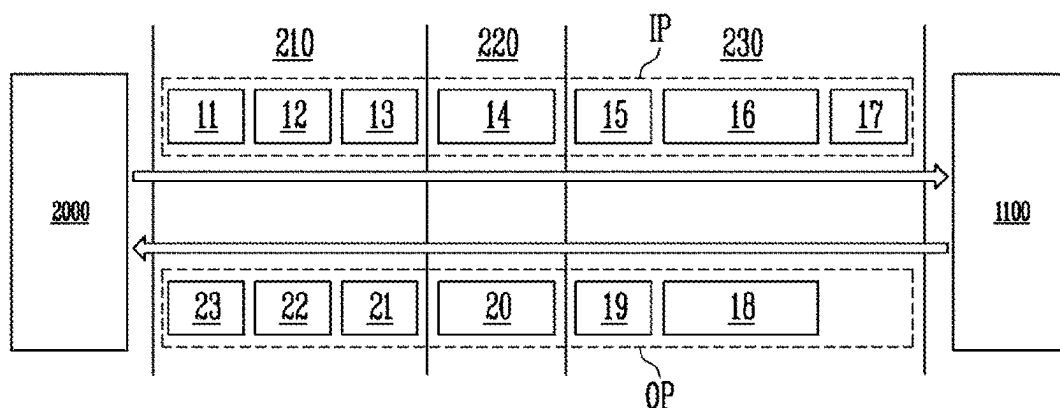


FIG. 4

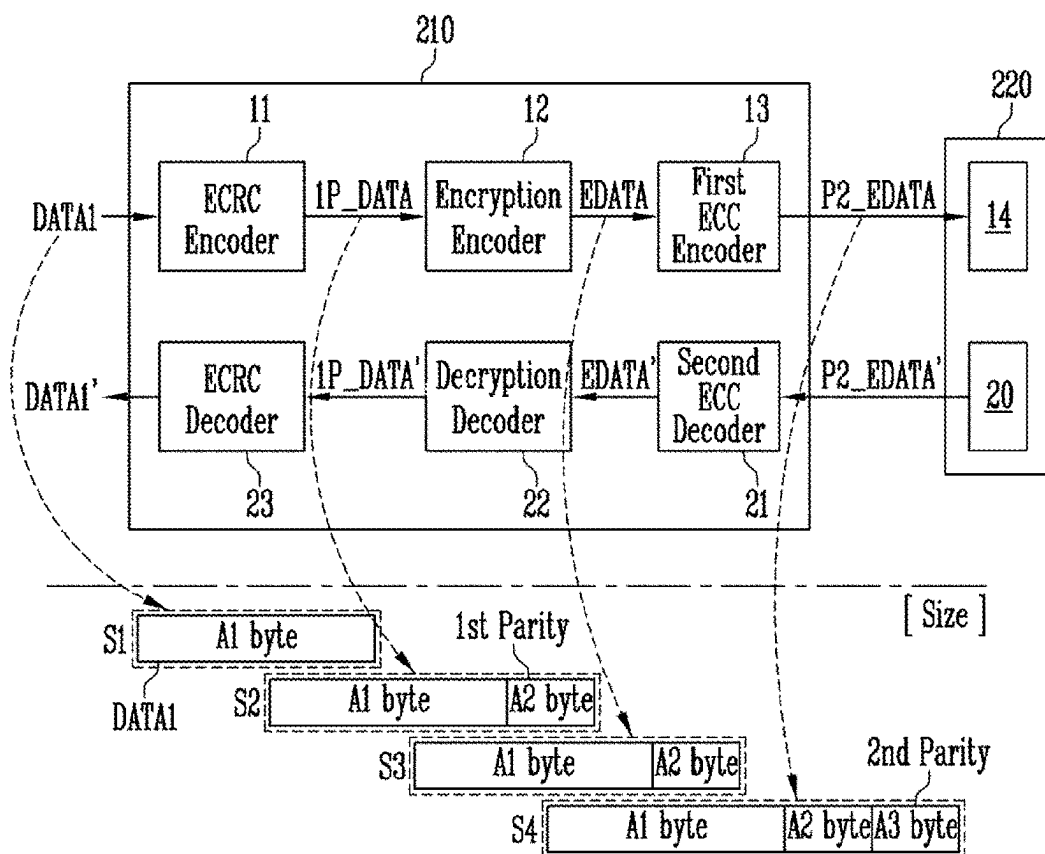


FIG. 5

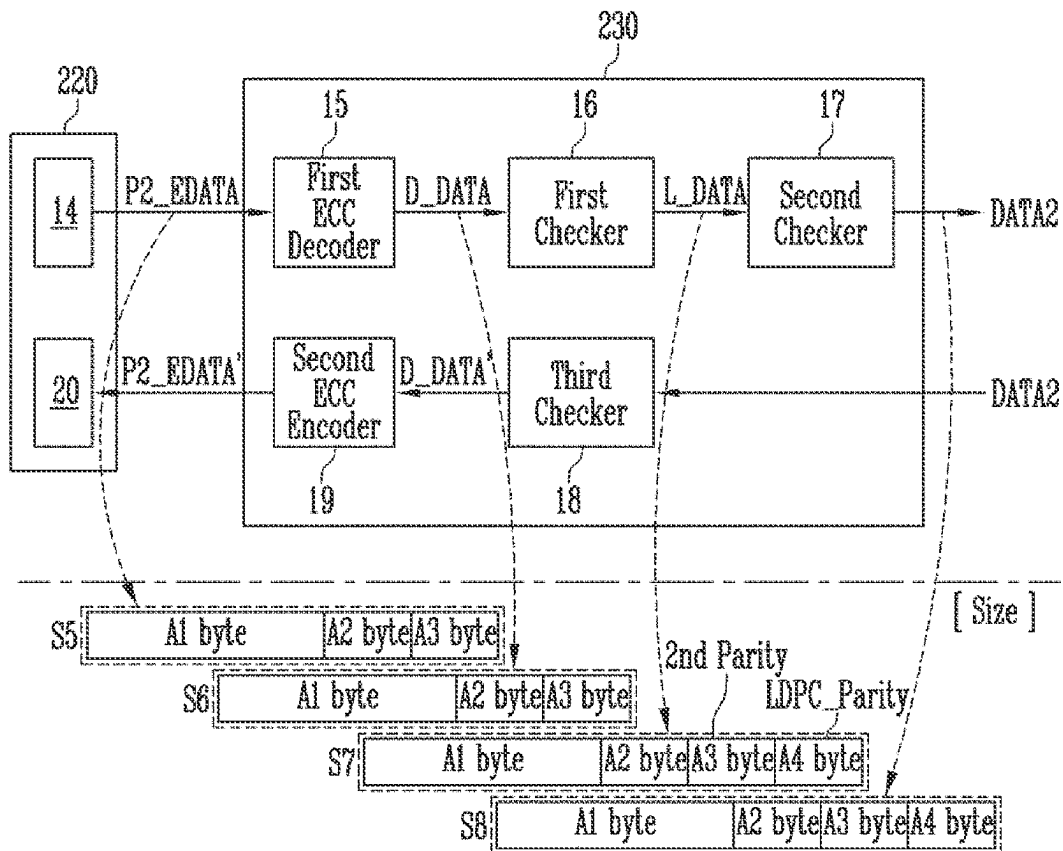


FIG. 6

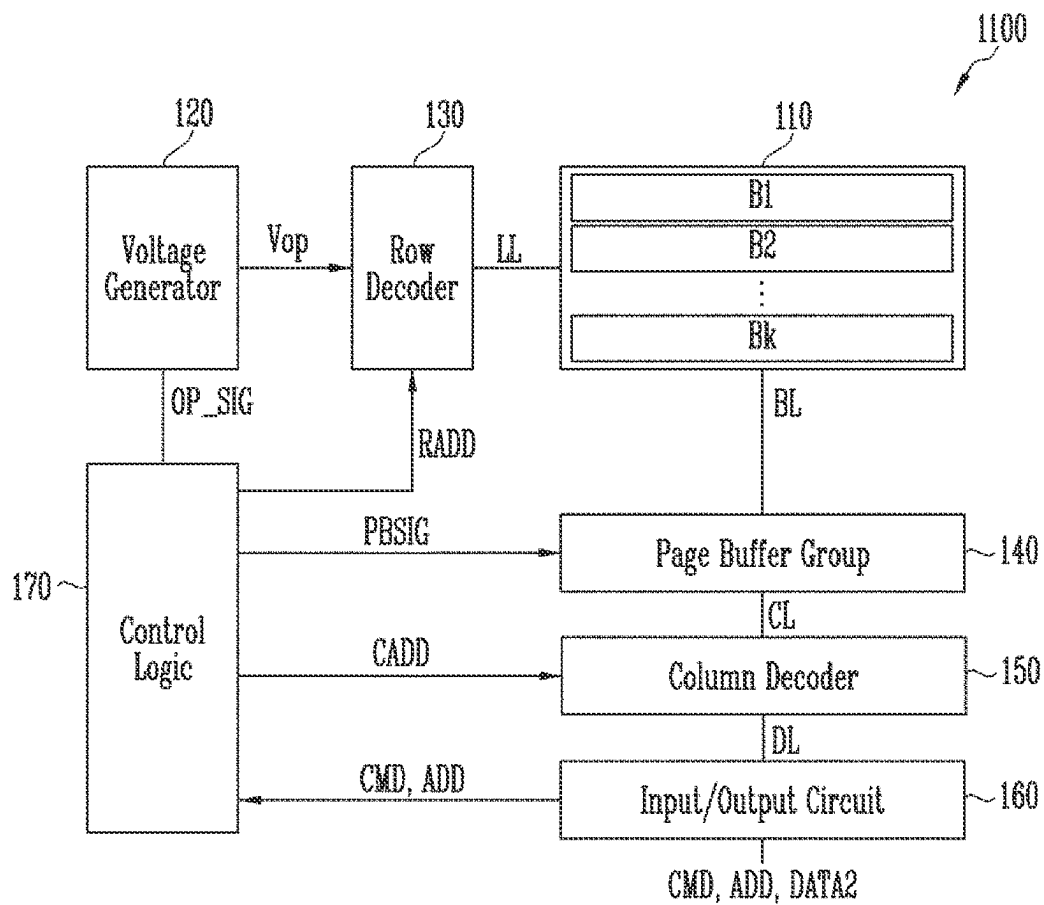


FIG. 7

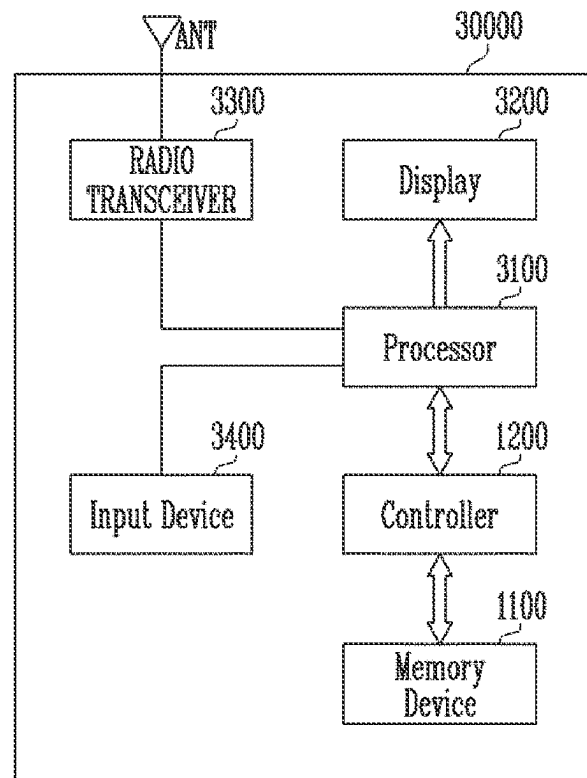


FIG. 8

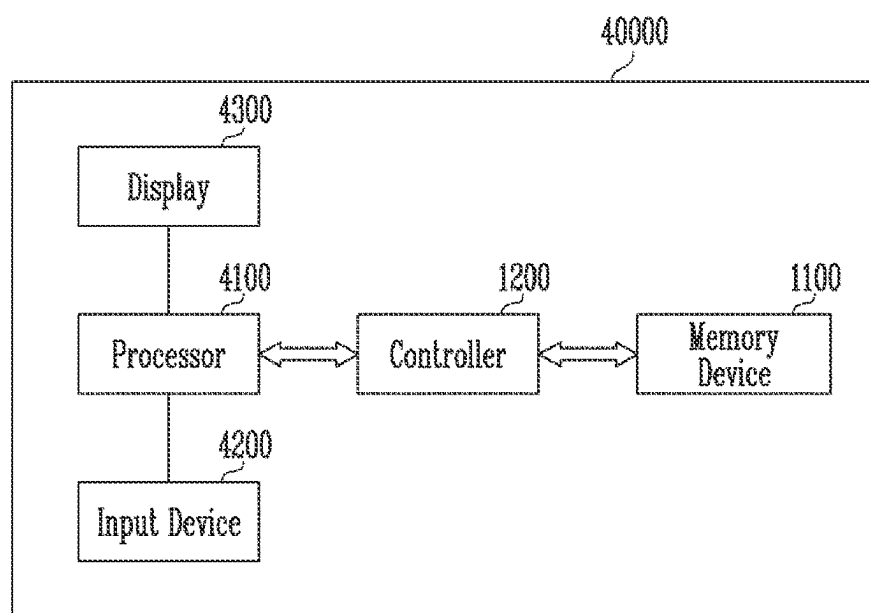


FIG. 9

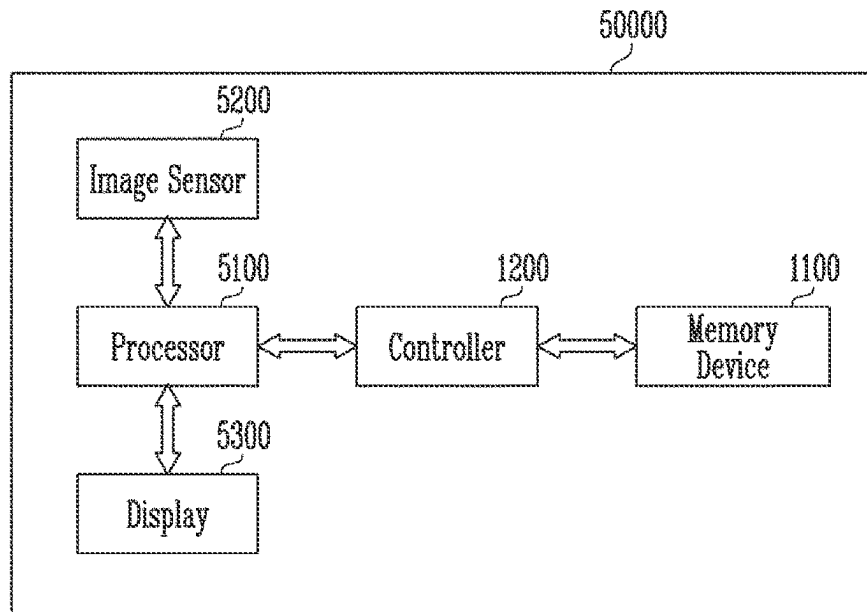
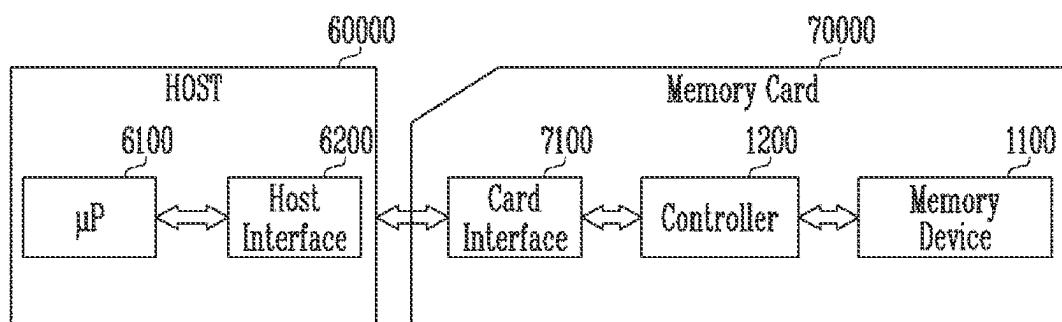


FIG. 10



CONTROLLER AND MEMORY SYSTEM**CROSS-REFERENCE TO RELATED APPLICATION**

This application is a continuation of U.S. patent application Ser. No. 17/851,130 filed on Jun. 28, 2022, which is a continuation of U.S. patent application Ser. No. 17/307,129 filed on May 4, 2021, and issued as U.S. Pat. No. 11,403,174 on Aug. 2, 2022, which is a division of U.S. patent application Ser. No. 16/555,264 filed on Aug. 29, 2019 and issued as U.S. Pat. No. 11,099,932 on Aug. 24, 2021, which claims benefits of priority of Korean Patent Application No. 10-2018-0168761 filed on Dec. 24, 2018. The disclosure of each of the foregoing applications is incorporated herein by reference in its entirety.

BACKGROUND**Field of Invention**

Various embodiments of the present disclosure generally relate to a controller and a memory system including the controller, and more particularly, to a data path protection technology of the controller.

Description of Related Art

A memory system includes a memory device for storing data and a controller for communicating between the memory device and a host.

The memory devices are generally classified into either volatile memory devices or non-volatile memory devices. Volatile memory devices may lose data stored therein when the power supply is blocked, and non-volatile memory devices may retain the stored data even in the absence of a power supply.

Recently, non-volatile memory devices have increasingly used as the use of portable electronic devices increases.

A controller may control operations of a non-volatile memory device, and transfer data between a host and the non-volatile memory device. A Data Path Protection (DPP) operation may be performed to increase the reliability of data to be transferred.

For example, an encryption operation, an encoding operation, a decoding operation, and/or an error check operation may be performed during the DPP operation. There is a demand to increase the reliability of each operation performed during the DPP operation while reducing an overall operation time.

SUMMARY

Various embodiments are directed to a controller capable of increasing the reliability of data with decreasing the time required for transferring the data during a write operation, and to a memory system including the controller.

According to an embodiment, a controller may include an Error Correction Code (ECC) encoder adding a first parity to data to generate a data set, and encoding the data set to generate a first parity data set; a buffer temporarily storing the first parity data set; an ECC decoder decoding the first parity data set received from the buffer to generate a decoded data set; a first checker performing a Low Density Parity Check (LDPC) encoding on the decoded data set to generate an LDPC data set to which a second parity is added; and a

second checker performing a syndrome check operation on the LDPC data set including the first and second parities.

According to an embodiment, a controller may include a host controller encrypting a first data which is received from a host, add a parity to the encrypted data, and encoding the encrypted data; a first buffer temporarily storing the encrypted data which is output from the host controller; and a memory controller adding an additional parity to the encrypted data, which is output from the first buffer, to perform an error check operation, wherein the memory controller outputs the error-checked data as a second data when no error is found.

According to an embodiment, a memory system may include a controller, during a write operation, adding first and second parities to original data which is received from a host, adding an LDPC parity to a first data set which includes the first and second parities, performing an LDPC encoding, and performing a syndrome check operation on a second data set, which includes the first, second, and LDPC parities; and a memory device storing the second data set, which is output from the controller, wherein, during a read operation, the controller receives the second data set from the memory device, performs an error correction operation on the second data set, and then outputs the original data to the host.

According to an embodiment, a data path protection method of a controller, the method includes encoding original data received from a host to generate encrypted data in which first and second parities are added; temporarily storing the encrypted data in a buffer; performing an LDPC encoding on the encrypted data received from the buffer to add an LDPC parity to generate a LDPC parity-added data; performing a syndrome check operation on the LDPC parity-added data including the first, second, and LDPC parities; and storing the LDPC parity-added data in a memory device.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a block diagram illustrating a memory system according to an embodiment of the present disclosure;

FIG. 2 is a detailed diagram of a controller included in the memory system shown in FIG. 1;

FIG. 3 is a detailed diagram of a data transfer path of the controller shown in FIG. 2;

FIG. 4 is a diagram for describing a data transfer method in a host controller included in the controller shown in FIG. 3;

FIG. 5 is a diagram for describing a data transfer method in a memory controller included in the controller shown in FIG. 3;

FIG. 6 is a detailed diagram of a memory device included in the memory system shown in FIG. 1; and

FIGS. 7 to 10 are block diagrams illustrating a memory system including a memory device and a controller, according to various embodiments of the present disclosure.

DETAILED DESCRIPTION

Advantages and features of the present invention and methods for achieving them will be made clear from embodiments described below in detail with reference to the accompanying drawings. However, they may be embodied in different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough

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and complete, and will fully convey the scope of the example embodiments to those skilled in the art.

It will be understood that when an element is referred to as being “coupled” or “connected” to a certain element, it may be “directly coupled or connected” to the certain element or may be “indirectly coupled or connected” to the certain element, with intervening elements being present therebetween.

It is noted that reference to “an embodiment,” “another embodiment” or the like does not necessarily mean only one embodiment, and different references to any such phrase are not necessarily to the same embodiment(s).

It will be understood that, although the terms “first”, “second”, “third”, and so on may be used herein to describe various elements, these elements are not limited by these terms. These terms are used to distinguish one element from another element. Thus, a first element described below could also be termed as a second or third element without departing from the spirit and scope of the present invention.

It will be further understood that the terms “comprises,” “comprising,” “includes,” and “including” when used in this specification, specify the presence of the stated elements and do not preclude the presence or addition of one or more other elements. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

As used herein, singular forms may include the plural forms as well and vice versa, unless the context clearly indicates otherwise. The articles ‘a’ and ‘an’ as used in this application and the appended claims should generally be construed to mean ‘one or more’ unless specified otherwise or clear from context to be directed to a singular form.

FIG. 1 is a block diagram illustrating a memory system 1000 according to an embodiment of the present disclosure.

Referring to FIG. 1, the memory system 1000 may include a memory device 1100 and a controller 1200.

The memory device 1100 may include a single storage device or a plurality of storage devices. Each storage device may include a plurality of memory cells storing data therein. The memory device 1100 may perform a write operation, a read operation, and an erase operation of data under the control of the controller 1200. The memory device 1100 may include at least one non-volatile memory device, e.g., NAND flash memory.

The controller 1200 may control the memory device 1100 in response to a request of a host 2000, and control the memory device 1100 through an internal operation without a request from the host 2000. For example, when the controller 1200 receives a write request and data from the host 2000, the controller 1200 may transfer the received data to the memory device 1100. The controller 1200 may not directly transfer the data received from the host 2000 to the memory device 1100, but may transfer the data to the memory device 1100 after performing an error check operation to increase the reliability of data. An error check operation performed in the controller 1200 will be described below.

The host 2000 may communicate with the memory device 1100 through the controller 1200 by using an interface protocol such as Peripheral Component Interconnect-Express (PCI-E), Advanced Technology Attachment (ATA), Serial ATA (SATA), Parallel ATA (PATA), or Serial Attached SCSI (SAS). The interface protocol used between the host 2000 and the memory system 1000 may not be limited to the above examples, and may include a Universal Serial Bus

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(USB), a Multi-Media Card (MMC), an Enhanced Small Disk Interface (ESDI), or Integrated Drive Electronics (IDE).

FIG. 2 is a detailed diagram of the controller 1200 included in the memory system 1000 shown in FIG. 1.

Referring to FIG. 2, the controller 1200 may transfer data received from the host 2000 shown in FIG. 1 to the memory device 1100 shown in FIG. 1, and transfer data read from the memory device 1100 to the host 2000. The controller 1200 includes a host controller 210, a buffer 220, and a memory controller 230. The host controller 210 may be referred to as host interface (I/F), and the memory controller 230 may be referred to as memory interface (I/F).

Hereinafter, data which is transferred from the host 2000 to the controller 1200 may be referred to as “first data DATA1”, and data which is transferred from the controller 1200 to the memory device 1100 may be referred to as “second data DATA2”. In other words, the first data DATA1 may be original data, and the second data DATA2 may be data converted from the first data DATA1.

The host controller 210, the buffer 220, and the memory controller 230 may convert the first data DATA1 into the second data DATA2, and transfer the second data DATA2 to the memory device 1100, along an input path IP. The host controller 210, the buffer 220, and the memory controller 230 may restore the second data DATA2 to the first data DATA1, and output the first data DATA1 to the host 2000, along an output path OP. In other words, the host controller 210, the buffer 220, and the memory controller 230 may include devices used in each of the input path IP and the output path OP, use the devices included in the input path IP during a write operation, and use the devices included in the output path OP during a read operation.

The host controller 210 may perform an encryption operation and a parity addition operation on the first data DATA1 during the write operation. During the read operation, the host controller 210 may perform a parity removal operation, restore the encrypted data to the first data DATA1, and transfer the first data DATA1 to the host 2000.

The buffer 220 may temporarily store the data transferred from the host controller 210 or the memory controller 230 during the write operation and the read operation.

The memory controller 230 may perform an error check operation on the data received from the buffer 220, and transfer the second data DATA2 on which the error check operation is performed to the memory device 1100, during the write operation. The memory controller 230 may perform an error check operation on the second data DATA2 received from the memory device 1100, and transfer the error-checked data to the buffer 220, during the read operation.

In other words, the devices included in the input path IP may be used to check whether there is communication failure of data which is transferred to the memory device 1100 through the controller 1200 during the write operation. Further, the devices included in the output path OP may be used to check whether there is communication failure of data which is output to the host 2000 through the controller 1200 during the read operation.

The host controller 210, the buffer 220, and the memory controller 230 as recited above will be described below in more detail.

FIG. 3 is a detailed diagram of a data transfer path of the controller shown in FIG. 2.

Referring to FIG. 3, the input path IP may include an external Cyclic Redundancy Check (CRC) encoder 11, an encryption encoder 12, a first Error Correction Code (ECC)

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encoder **13**, a first buffer **14**, a first Error Correction Code (ECC) decoder **15**, a first checker **16**, and a second checker **17**. The output path OP may include a third checker **18**, a second ECC encoder **19**, a second buffer **20**, a second ECC decoder **21**, a decryption decoder **22**, and an external Cyclic Redundancy Check (CRC) decoder **23**.

The external CRC (ECRC) encoder **11**, the encryption encoder **12**, the first ECC encoder **13**, the second ECC decoder **21**, the decryption decoder **22**, and the external CRC (ECRC) decoder **23** may be included in the host controller **210**. The first and second buffers **14** and **20** may be included in the buffer **220**. The first ECC decoder **15**, the first checker **16**, the second checker **17**, the third checker **18**, and the second ECC encoder **19** may be included in the memory controller **230**.

Accordingly, the external CRC encoder **11**, the encryption encoder **12**, and the first ECC encoder **13** included in the input path IP of the host controller **210**, the first buffer **14** included in the input path IP of the buffer **220**, the first ECC decoder **15**, the first checker **16**, and the second checker **17** included in the input path IP of the memory controller **230** may operate during the write operation. The third checker **18** and the second ECC encoder **19** included in the output path OP of the memory controller **230**, the second buffer **20** included in the output path OP of the buffer **220**, the second ECC decoder **21**, the decryption decoder **22**, and the external CRC decoder **23** included in the output path OP of the host controller **210** may operate during the read operation.

Each of the devices included in the host controller **210**, the buffer **220**, and the memory controller **230** as recited above will be described below in more detail.

FIG. 4 is a diagram for describing a data transfer method in the host controller **210** included in the controller **1200** shown in FIG. 3. FIG. 5 is a diagram for describing a data transfer method in the memory controller **230** included in the controller **1200** shown in FIG. 3. FIG. 6 is a detailed diagram of the memory device **1100** included in the memory system **1000** shown in FIG. 1.

Each of the write operation and the read operation will be described with reference to FIGS. 4, 5, and 6 as below.

Write Operation

Referring to FIG. 4, the write operation may refer to an operation of converting the first data DATA1 received from the host **2000** into the second data DATA2 and transferring the second data DATA2 to the memory device **1100** when a write request is provided from the host **2000**. This write operation may be performed in the controller **1200**.

During the write operation, when receiving the first data DATA1 from the host **2000**, the ECRC encoder **11** may output a first parity data set 1P_DATA, that is, the first data DATA1 to which a first parity 1st Parity is added. The first parity 1st Parity may be added to the first data DATA1 to encrypt the first data DATA1. For example, it may be assumed that a size "S1" of the first data DATA1 is "A1 byte", and a size "S2" of the first parity data set 1P_DATA including the first parity 1st Parity of A2 byte is "A1+A2 byte". For example, a size of parity may be about 10% of a size of data.

The encryption encoder **12** may encrypt the first parity data set 1P_DATA to output an encrypted data set EDATA. The encryption encoder **12** may encrypt the first parity data set 1P_DATA by using a Data Encryption Standard (DES) algorithm or an Advanced Encryption Standard (AES) algorithm. The AES algorithm has the improved encryption performance compared to the DES algorithm. Even when the first parity data set 1P_DATA is encrypted, the number of entire bits may not be changed. Accordingly, a size S3 of

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the encrypted data set EDATA may be the same as the size S2 of the first parity data set 1P_DATA. In some cases, the size S3 of the encrypted data set EDATA may be different from the size S2 of the first parity data set 1P_DATA. Since the first data DATA1 and the first parity 1st Parity included in the first parity data set 1P_DATA are encrypted by the encryption encoder **12**, the first data DATA1 and the first parity 1st Parity may not be distinguished from each other in the encrypted data set EDATA.

The first ECC encoder **13** may add a second parity 2nd Parity to the encrypted data set EDATA, encode the encrypted data set EDATA and the second parity 2nd Parity to output a second parity data set P2_EDATA. The second parity 2nd Parity may have a size of A3 byte. Accordingly, the size of the second parity 2nd Parity may be added to the size S3 of the encrypted data set EDATA, so that a size S4 of the second parity data set P2_EDATA may be A1+A2+A3 byte.

The second parity data set P2_EDATA may be temporarily stored in the first buffer **14** before performing Low Density Parity Check (LDPC) encoding.

The first buffer **14** may temporarily store the second parity data set P2_EDATA until the write operation is completed. The first buffer **14** may be provided as a volatile memory or a non-volatile memory. For example, the first buffer **14** may be embodied as Dynamic Random Access Memory (DRAM), Static Random Access Memory (SRAM), or a non-volatile memory.

Referring to FIG. 5, the first ECC decoder **15** may decode the second parity data set P2_EDATA which is temporarily stored in the first buffer **14** to output a decoded data set D_DATA. A size S5 of the second parity data set P2_EDATA input to the first ECC decoder **15** may be the same as the size S4 of the second parity data set P2_EDATA input to the first buffer **14**.

The decoded data set D_DATA may be the same as the encrypted data set EDATA input to the first ECC encoder **13**. For example, since the second parity data set P2_EDATA which is encoded in the first ECC encoder **13**, is decoded by the first ECC decoder **15**, a value of the decoded data set D_DATA output from the first ECC decoder **15** may be the same as a value of the encrypted data set EDATA before being encoded by the first ECC encoder **13**. Accordingly, a size S6 of the decoded data set D_DATA may be the same as the size S5 of the second parity data set P2_EDATA.

The first checker **16** may encode the decoded data set D_DATA by Low Density Parity Check (LDPC) encoding to output an LDPC data set L_DATA. For example, the first checker **16** may perform an LDPC encoding operation by multiplying the decoded data set D_DATA by a generation matrix, and output the LDPC data set L_DATA generated by a result of the operation.

$$[D_{DATA} < 1 > \dots D_{DATA} < N >] \cdot \begin{bmatrix} A_{11} & \dots & A_{1N} \\ \vdots & \ddots & \vdots \\ A_{K1} & \dots & A_{KN} \end{bmatrix} = [L_{DATA} < 1 > \dots L_{DATA} < K >] \quad [\text{Equation 1}]$$

Referring to 'Equation 1', the generation matrix may include rows of 'K' bits (or bytes) and columns of 'N' bits (or bytes), and may be differently designed according to LDPC performance. In other words, a single row may include 'N' bits (or bytes). Here, both 'K' and 'N' are positive integers and 'N' is greater than 'K'. 'K' may be the same as the size S6 of the decoded data set D_DATA.

Accordingly, a size **S7** of the LDPC data set **L_DATA** generated by the LDPC encoding operation may be 'N' bits (or bytes), and an LDPC parity **LDPC_Parity** may be added as much as difference between 'K' and 'N'. In other words, the LDPC parity **LDPC_Parity** of (N-K) bits (or bytes) may be added to the size **S6** of the decoded data set **D_DATA** to result in the size **S7** of the LDPC data set **L_DATA**. For example, when 'K' is **A1+A2+A3** bytes and 'N' is **A1+A2+A3+A4** bytes, the LDPC parity **LDPC_Parity** added to the LDPC data set **L_DATA** may be **A4** bytes. The size of the LDPC parity **LDPC_Parity** may change according to a generation matrix set in the first checker **16**. The LDPC data set **L_DATA** may be a code word generated by a result of LDPC encoding. **A2**, **A3**, and **A4** may be the same as or different from each other.

The second checker **17** may perform a syndrome check operation on the LDPC data set **L_DATA**. For example, the second checker **17** may calculate a syndrome vector (**S** < (N-K):1 >) of (N-K) bits (or bytes) by multiplying a parity check matrix by the LDPC data set **L_DATA**. The second checker **17** may determine whether a syndrome check operation passes or fails according to the syndrome vector (**S** < (N-K):1 >).

$$\begin{bmatrix} B_{11} & \cdots & B_{1N} \\ \vdots & \ddots & \vdots \\ B_{(N-K)1} & \cdots & B_{(N-K)N} \end{bmatrix} \cdot \begin{bmatrix} \text{L_DATA} < 1 > \\ \vdots \\ \text{L_DATA} < N > \end{bmatrix} = \begin{bmatrix} \text{S} < 1 > \\ \vdots \\ \text{S} < N - K > \end{bmatrix} \quad [\text{Equation 2}]$$

Referring to 'Equation 2', a parity check matrix may include rows of (N-K) bits (or bytes) and columns of N bits (or bytes). In other words, a single row may include N bits (or bytes). The size **S7** of the syndrome vector (**S** < (N-K):1 >) calculated by multiplying the parity check matrix by the LDPC data set **L_DATA** may be (N-K) bits (or bytes). When all bits included in the syndrome vector (**S** < (N-K):1 >) are '0', the syndrome check operation passes. When '1' is included in the bits included in the syndrome vector (**S** < (N-K):1 >), the syndrome check operation fails.

The second checker **17** may perform the syndrome check operation of the LDPC data set **L_DATA** by changing check nodes until all bits included in the syndrome vector (**S** < (N-K):1 >) become '0'. In other words, the second checker **17** may perform the syndrome check operation on a modified data set which includes the first data **DATA1**, the first parity 1st Parity, the second parity 2nd Parity, and the LDPC parity **LDPC_Parity**.

When the syndrome check operation passes, the second checker **17** may output the second data **DATA2**. The second data **DATA2** may be the LDPC data set **L_DATA** on which the syndrome check operation passes. The second data **DATA2** output from the second checker **17** may be transferred to the memory device **1100**, and the memory device **1100** may program the received second data **DATA2** into a selected memory region.

When the syndrome check operation fails, the second checker **17** may search for a check node value to make all bits included in the syndrome vector (**S** < (N-K):1 >) become '0'.

An embodiment of the memory device **1100** will be described in detail with reference to FIG. 6 as below.

Referring to FIG. 6, the memory device **1100** may include a memory cell array **110** for storing data, peripheral circuits

for performing program, read or erase operations, and a control logic **170** for controlling the peripheral circuits based on a command **CMD** and an address **ADD**. For example, the peripheral circuits may include a voltage generator **120**, a row decoder **130**, a page buffer group **140**, a column decoder **150**, and an input/output circuit **160**.

The memory cell array **110** may include a plurality of memory blocks **B1** to **Bk**, where **k** is a positive integer. The memory blocks **B1** to **Bk** may include a plurality of memory cells and have a two-dimensional or three-dimensional structure. For example, in two-dimensionally structured memory blocks **B1** to **Bk**, memory cells may be arranged in a horizontal direction to a substrate. In three-dimensionally structured memory blocks **B1** to **Bk**, memory cells may be stacked in a vertical direction to a substrate.

The voltage generator **120** may generate and output operating voltages **Vop** necessary for respective operations in response to operating signals **OP_SIG**. For example, when the operating signals **OP_SIG** for the program operation are provided, the voltage generator **120** may generate a program voltage, a pass voltage, a program verify voltage, and the like. When the operating signals **OP_SIG** for the read operation are provided, the voltage generator **120** may generate a read voltage, a pass voltage, and the like. When the operating signals **OP_SIG** for the erase operation are provided, the voltage generator **120** may generate an erase voltage, a pass voltage, an erase verify voltage, and the like.

The row decoder **130** may transfer the operating voltages **Vop** to a selected memory block through local lines **LL** according to a row address **RADD**.

The page buffer group **140** may be coupled to the memory blocks **B1** to **Bk** through bit lines **BL**, and include page buffers coupled to the bit lines **BL**, respectively. The page buffer group **140** may control a voltage of the bit lines **BL** or sense a voltage or a current of the bit lines **BL** in response to page buffer control signals **PBSIG**.

The column decoder **150** may exchange data with the page buffer group **140**, or may exchange data with the input/output circuit **160** in response to a column address **CADD**. For example, the column decoder **150** may exchange data with the page buffer group **140** through column lines **CL**, and exchange data with the input/output circuit **160** through data lines **DL**.

The input/output circuit **160** may receive the command **CMD**, an address **ADD**, and the second data **DATA2** from the controller **1200** during the write operation, and output the second data **DATA2** to the controller **1200** during the read operation. For example, the input/output circuit **160** may transfer the received command **CMD** and address **ADD** to the control logic **170**, and transfer the second data **DATA2** to the column decoder **150**. In addition, the input/output circuit **160** may output the second data **DATA2** read from the memory blocks **B1** to **Bk** to the controller **1200**.

The control logic **170** may output control signals for controlling the peripheral circuits based on the command **CMD** and the address **ADD**. The control signals may include the operating signals **OP_SIG** and the page buffer control signals **PBSIG**. In addition, the control logic **170** may output the row address **RADD** and the column address **CADD** in response to the address **ADD**.

As described above, the data check operation is performed on the data set, that is, the first data **DATA1**, which is provided from the host **2000** during the write operation, to which the first and second parities 1st Parity and 2nd Parity and the LDPC parity **LDPC_Parity** are added. Accordingly, the logic for transferring data may be simplified when compared to the conventional method using more parities

than the parities as described above. In addition, the reliability of a data transfer method may be improved without increasing the number of buffers used for transferring data in the controller **1200**.

Read Operation

Referring to FIG. **5**, the read operation may refer to an operation of converting the second data **DATA2** read from the memory device **1100** into the first data **DATA1** and outputting the first data **DATA1** to the host **2000** when a read request is provided from the host **2000**. This read operation may be performed in the controller **1200**.

When a second data **DATA2'** read from the memory device **1100** is received, the third checker **18** may perform an error correction operation on the second data **DATA2** with removing the LDPC parity **LDPC_Parity** included in the second data **DATA2** to output a data set **D_DATA'**.

The second ECC encoder **19** may encode the data set **D_DATA'** output from the third checker **18** to output an encoded data set **P2_EDATA'**. The encoded data set **P2_EDATA'** output from the second ECC encoder **19** may be temporarily stored in the second buffer **20** included in the buffer **220**.

The second buffer **20** may temporarily store the encoded data set **P2_EDATA'** until the read operation is completed. The second buffer **20** may be provided as a volatile memory or a non-volatile memory. For example, the second buffer **20** may be embodied as Dynamic Random Access Memory (DRAM), Static Random Access Memory (SRAM), or flash memory.

Referring back to FIG. **4**, the second ECC decoder **21** may decode the encoded data set **P2_EDATA'** stored in the second buffer **20** to output a decoded data set **EDATA'**. The decoded data set **EDATA'** output from the second ECC decoder **21** may be the same data as the encrypted data set **EDATA** output from the encryption encoder **12**. The second ECC decoder **21** may remove the second parity **2nd Parity** from the encoded data set **P2_EDATA'** while decoding, and output the decoded data set **EDATA'**.

The decryption decoder **22** may decrypt the decoded data set **EDATA'** to output a decrypted data set **1P_DATA'** which is decrypted.

The ECRC decoder **23** may decode the decoded data set **1P_DATA'** with removing the first parity **1st Parity** from the decoded data set **1P_DATA'** to output a first data **DATA1'**. In other words, the first data **DATA1'** output from the ECRC decoder **23** may be output to the host **2000**.

FIG. **7** is a diagram illustrating a memory system **30000** including a memory device and a controller according to an embodiment of the present disclosure.

Referring to FIG. **7**, the memory system **30000** may be embodied into a cellular phone, a smartphone, a tablet PC, a personal digital assistant (PDA), or a wireless communication device. The memory system **30000** may include the memory device **1100** and the controller **1200** controlling the operations of the memory device **1100**. The controller **1200** may control a data access operation, for example, a program operation, an erase operation, or a read operation of the memory device **1100** in response to control of a processor **3100**.

The controller **1200** may control data programmed into the memory device **1100** to be output through a display **3200**.

A radio transceiver **3300** may exchange a radio signal through an antenna **ANT**. For example, the radio transceiver **3300** may change the radio signal received through the antenna **ANT** into a signal which can be processed by the processor **3100**. Accordingly, the processor **3100** may pro-

cess the signal output from the radio transceiver **3300** and transfer the processed signal to the controller **1200** or the display **3200**. The controller **1200** may transfer the signal processed by the processor **3100** to the memory device **1100**. In addition, the radio transceiver **3300** may change a signal output from the processor **3100** into a radio signal and output the radio signal to an external device through the antenna **ANT**. A control signal for controlling the operations of the processor **3100** or data to be processed by the processor **3100** may be input by an input device **3400**, and the input device **3400** may be embodied into a pointing device, such as a touch pad and a computer mouse, a keypad, or a keyboard. The processor **3100** may control operations of the display **3200** so that the data output from the controller **1200**, the data output from the radio transceiver **3300**, or the data output from the input device **3400** are displayed on the display **3200**.

According to an embodiment, the controller **1200** for controlling the operations of the memory device **1100** may be formed as a part of the processor **3100**, or a separate chip from the processor **3100**.

FIG. **8** is a diagram illustrating a memory system **40000** including a memory device and a controller according to an embodiment of the present disclosure.

Referring to FIG. **8**, the memory system **40000** may be embodied into a personal computer (PC), a tablet PC, a net-book, an e-reader, a personal digital assistant (PDA), a portable multimedia player (PMP), an MP3 player, or an MP4 player.

The memory system **40000** may include the memory device **1100** storing data and the controller **1200** controlling a data processing operation of the memory device **1100**.

A processor **4100** may output data stored in the memory device **1100** through a display **4300** according to data input through an input device **4200**. Examples of the input device **4200** may include a pointing device such as a touch pad or a computer mouse, a keypad, or a keyboard.

The processor **4100** may control the general operations of the memory system **40000** and control the operations of the controller **1200**. According to an embodiment, the controller **1200** for controlling the operations of the memory device **1100** may be formed as a part of the processor **4100**, or a separate chip from the processor **4100**.

FIG. **9** is a diagram illustrating a memory system **50000** including a memory device and a controller according to an embodiment of the present disclosure.

Referring to FIG. **9**, the memory system **50000** may be embodied into an image processor, for example, a digital camera, a smartphone with a digital camera attached thereto, or a tablet PC with a digital camera attached thereto.

The memory system **50000** may include the memory device **1100** and the controller **1200** controlling a data processing operation of the memory device **1100**, for example, a program operation, an erase operation, or a read operation.

An image sensor **5200** of the memory system **50000** may convert an optical image into digital signals, and the converted digital signals may be transferred to a processor **5100** or the controller **1200**. In response to control of the processor **5100**, the digital signals may be output through a display **5300** or stored in the memory device **1100** through the controller **1200**. In addition, the data stored in the memory device **1100** may be output through the display **5300** according to control of the processor **5100** or the controller **1200**.

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According to an embodiment, the controller **1200** for controlling the operations of the memory device **1100** may be formed as a part of the processor **5100**, or a separate chip from the processor **5100**.

FIG. **10** is a diagram illustrating a memory system **70000** including a memory device and a controller according to an embodiment of the present disclosure.

Referring to FIG. **10**, the memory system **70000** may be embodied into a memory card (or a smart card). The memory system **70000** may include the memory device **1100**, the controller **1200** and a card interface **7100**.

The controller **1200** may control data exchange between the memory device **1100** and the card interface **7100**. According to an embodiment, the card interface **7100** may be, but not limited to, a secure digital (SD) card interface or a multi-media card (MMC) interface.

The card interface **7100** may interface data exchange between a host **60000** and the controller **1200** according to a protocol of the host **60000**. According to an embodiment, the card interface **7100** may support a Universal Serial Bus (USB) protocol and an InterChip (IC)-USB protocol. The card interface **7100** may refer to hardware that supports a protocol used by the host **60000**, software mounted on the hardware, or a signal transmission method.

When the memory system **70000** is connected to an host interface **6200** of the host **60000** such as a PC, a tablet PC, a digital camera, a digital audio player, a cellular phone, console video game hardware, or a digital set-top box, the host interface **6200** may perform data communication with the memory device **1100** through the card interface **7100** and the controller **1200** in response to control of a microprocessor (μ P) **6100**.

As described above, according to embodiments of the present disclosure, during a write operation of data, logic for transferring data within a controller may be simplified and the reliability of a data transfer operation may be improved without increasing the number of buffers used to transfer data.

Examples of embodiments have been disclosed herein, and although specific terms are employed, various changes in forms and details may be made to the above-described examples of embodiments without departing from the spirit and scope of the present invention. Accordingly, it will be understood by those skilled in the art that the scope of the present invention should not be limited to the above-described examples of embodiments, and may cover the scope of the following claims and their equivalents.

What is claimed is:

1. An operating method for a controller of a memory device, the method comprising:

receiving, from a host, a write request of data;
adding a parity to the data to obtain a first data set including the parity;
performing an ecc encoding operation to obtain a data set by adding an ecc parity to the first data set;
performing an ecc decoding operation on the data set;
performing an ldpc encoding operation by adding an ldpc parity to the first data set to obtain a second data set which includes the parity and the ldpc parity; and
transferring, to the memory device, the second data set according to a result of a syndrome check operation using the ecc parity on the second data set.

2. The method according to claim 1, wherein the transferring the second data set comprises transferring the second data set when the syndrome check operation on the second data set passes.

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3. The method according to claim 1, wherein the transferring the second data set comprises searching for a check node value to make the syndrome check operation pass when the syndrome check operation on the second data set fails.

4. The method according to claim 1, further comprising: storing the data set in a buffer.

5. The method according to claim 4, further comprising: maintaining the data set in the buffer until the second data set is stored in the memory device.

6. The method according to claim 4, wherein the buffer includes at least one of a volatile memory or a non-volatile memory.

7. The method according to claim 1, further comprising: encrypting the first data set using a Data Encryption Standard (DES) algorithm or an Advanced Encryption Standard (AES) algorithm.

8. The method according to claim 1, wherein the parity is used for a cyclic redundancy check of data.

9. The method according to claim 1, wherein the first data set is composed of a first bit number and the second data set is composed of a second bit number greater than the first bit number.

10. An operating method for a controller of a memory device, the method comprising:

receiving, from a host, a write request of data;
producing sets of data including a parity and a ldpc parity by performing an ldpc encoding operation;
transferring the sets of data to the memory device;
receiving, from the host, a read request of the data;
receiving, from the memory device, the sets of data;
restoring the data by performing an ldpc decoding operation;

transferring the data to the host,
wherein the producing sets of data comprises:
adding a parity to the data to obtain a first data set;

adding an ecc parity to the first data set to obtain a data set;
decoding the data set to remove the ecc parity from the data set;

encoding the data set to obtain a second data set which includes the parity and the ldpc parity; and
checking the second data set by performing a syndrome check operation using the ecc parity.

11. The method according to claim 10, wherein the producing sets of data further comprises storing the data set in a buffer and maintaining the data set in the buffer until the sets of data are stored in the memory device.

12. The method according to claim 11, wherein the producing sets of data further comprises encrypting the first data set using a Data Encryption Standard (DES) algorithm or an Advanced Encryption Standard (AES) algorithm.

13. The method according to claim 10, wherein the parity is used for a cyclic redundancy check of data.

14. The method according to claim 10, wherein the first data set is composed of a first bit number and the second data set is composed of a second bit number greater than the first bit number.

15. The method according to claim 10, wherein the restoring the data by performing the ldpc decoding operation comprises:
performing the ldpc decoding operation on the sets of data to obtain decoded data by removing the ldpc parity;
performing an ecc encoding operation to obtain a data set by adding an ecc parity to the decoded data;

performing an ecc decoding operation on the data set by removing the ecc parity; and performing a decoding operation to obtain the data by removing the parity from the decoded data.

16. The method according to claim **15**, 5

wherein the restoring the data further comprises storing the data set in a buffer and maintaining the data set in the buffer until the transferring the data to the host is completed.

17. The method according to claim **15**, 10

wherein the restoring the data further comprises decrypting the decoded data using a Data Encryption Standard (DES) algorithm or an Advanced Encryption Standard (AES) algorithm.

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